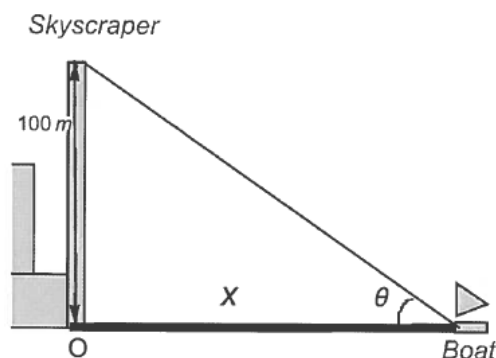


Problem 10

Problem 10

Problem 1 A boat sails directly toward a 100-meter skyscraper that stands on the edge of a harbor. The angular size θ of the building is the angle formed by lines from the top and bottom of the building to the observer on the boat (see figure below).



- (a) Express the angle θ as the function of x , the distance of the boat from the building.

Explanation. $\tan \theta = \frac{100}{x} \implies \theta = \tan^{-1} \left(\frac{100}{x} \right)$

- (b) The boat is sailing directly toward the skyscraper at 3 m/s. Find $\frac{d\theta}{dt}$ when the boat is $x = 300\text{m}$ from the building. [3]

Explanation.

$$\begin{aligned}
 \frac{d\theta}{dt} &= \frac{d}{dt} \tan^{-1} \left(\frac{100}{x} \right) \\
 &= \frac{d}{dx} \tan^{-1} \left(\frac{100}{x} \right) \frac{dx}{dt} \\
 &= \frac{1}{1 + \left(\frac{100}{x} \right)^2} \cdot \frac{-100}{x^2} \frac{dx}{dt} \\
 &= \frac{1}{1 + \left(\frac{100}{x} \right)^2} \cdot \frac{-100}{x^2} \cdot (-3) \\
 &= \frac{300}{x^2 + 100^2}
 \end{aligned}$$

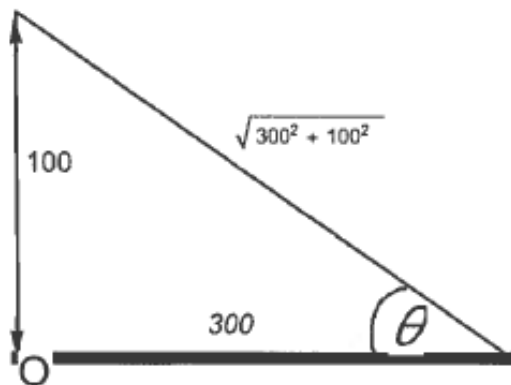
$$\begin{aligned}
 \left[\frac{d\theta}{dt} \right]_{x=300} &= \frac{300}{300^2 + 100^2} \\
 &= \frac{3}{900 + 100} \\
 &= \frac{3}{1000} \\
 &= 0.003 \text{ rad/sec}
 \end{aligned}$$

Alternatively, this problem could be done without inverse functions.

$$\begin{aligned}
 \tan \theta &= \frac{100}{x} \\
 \frac{d}{dt}(\tan \theta) &= \frac{d}{dt} \left(\frac{100}{x} \right) \\
 \sec^2 \theta \frac{d\theta}{dt} &= \frac{-100}{x^2} \frac{dx}{dt}
 \end{aligned}$$

We have to evaluate $\sec^2 \theta$ at the moment when $x = 300\text{m}$

Problem 10



Using the triangle above, we have:

$$\sec \theta = \frac{\sqrt{300^2 + 100^2}}{300} = 100 \frac{\sqrt{3^2 + 1^2}}{300} = \frac{\sqrt{10}}{3}, \text{ we obtain:}$$

$$\begin{aligned} \left(\frac{\sqrt{10}}{3} \right)^2 \left[\frac{d\theta}{dt} \right]_{x=300} &= -\frac{100}{300^2} (-3) \\ \left[\frac{d\theta}{dt} \right]_{x=300} &= \frac{1}{300} \cdot \frac{9}{10} = \frac{3}{1000} \text{ rad/sec} \end{aligned}$$